Approach Note: Home Credit LAP Qualification Voice Agent

Role: AI Intern (Prompt Engineer) assignment | Prepared by: Om | Platform: Bolna AI

1. Objective

Build an outbound AI voice agent for Home Credit that informs existing customers about a pre-approved Loan Against Property (LAP) offer of up to 75 lakh rupees, verifies the customer, collects seven eligibility data points, applies the business rules exactly, and hands qualified leads to a senior loan expert. The agent is a first-contact qualifier, not a decision-maker. Success means rule accuracy, memory of what the customer already said, and a natural, advisory voice experience.

2. Business rules converted into an explicit rule table

Vague instructions such as "check eligibility" invite inconsistent behaviour, so every criterion in the assignment is written as an exact condition inside the prompt.

#	Data point	Eligible	Not eligible / special handling
1	Property type	Residential, Commercial, Industrial	Agricultural: immediate disqualification
2	Ownership	Sole or Joint	Both eligible; never disqualify joint ownership
3	Original documents	Originals available for verification	Photocopies only or none: disqualify
4	Loan amount	Up to 75,00,000	Above the cap: do not reject; explain the limit and ask whether to proceed with 75 lakh
5	Occupation + income	Salaried or Self-employed, income via Bank	Cash income: disqualify (one checklist item holding two values)
6	Market value	Any customer estimate	No minimum threshold exists in the brief, so none is invented
7	Tenure	3 to 15 years inclusive	Below 3 or above 15: disqualify

Two global switches sit above the checklist: a **busy customer** gets a callback time and the call ends, and an **existing loan or EMI-reduction** request goes to the loan-transfer specialist message and the call ends, with no checklist questions asked.

3. Prompt architecture and state design

The prompt is organised into sections that mirror the call: identity and demeanour, conversation starter (identify as an AI, ask for the right person), conversation flow, closing lines, and approved FAQ answers. All eleven assignment variables (company, customer, agent, gender, date, day, time, RAG context, language, history, utterance) are used as placeholders; nothing is hardcoded. Gender and language are enforced explicitly, including gendered verb forms for Hindi.

The model is told to track a silent internal state (property type, ownership, documents, loan amount, occupation, income mode, market value, tenure, callback time, existing-loan flag, disqualified) and never to speak it. On every customer turn it follows one fixed procedure:

1. **Extract** every checklist fact in the message, including facts nobody asked for yet.
2. **Overwrite** earlier values if the customer corrects themselves (the latest clear value wins).
3. **Check the transfer switch** (existing loan, EMI reduction, originals held by a lender).
4. **Check disqualifiers** using the latest values; if one fails, close politely and end the call.
5. **Check the loan cap** and handle amounts above 75 lakh.
6. **Hand off** only when all seven items are answered and every rule has passed.
7. Otherwise **ask only the earliest unanswered item**, never repeating anything already answered.

This "extract, validate, update, ask" order is what makes out-of-order answers (for example "residential house, about 1.2 crore, jointly owned with my wife") work: three items are captured in one turn and the agent moves straight to the next missing question. Optional depth follow-ups (city, documents held, employer or business type) are limited to one per item, are treated only as notes for the loan expert, and can never cause disqualification.

4. Voice-specific design

- Short turns (one or two sentences), one question at a time, no lists read aloud, amounts spoken naturally (one crore equals 100 lakh; months converted to years).
- Tolerance for speech-to-text noise: fillers are ignored, corrections overwrite, interruptions are answered briefly before returning to the checklist, unclear answers trigger one clarification rather than an assumption.
- Honest identity: the agent says it is an AI assistant at the start and asks for the customer by name. If someone else answers, no offer details are shared.
- Guardrails: no invented interest rates, fees, branches or minimum property values; product answers come only from the RAG variable, otherwise the senior expert will explain; never say "approved"; never ask for OTP, ID numbers or bank details.

5. Judgment calls on points the brief leaves open

Situation	Decision	Reason
Tenure above 15 years	Disqualify	The brief states the range as 3 to 15 and defines out-of-range answers as disqualifying; only the loan amount has a counter-offer rule.
Mixed bank and cash income	Ask where the main income arrives; decide on that	Brief requires income via bank but does not cover partial cash.
Originals held by a bank or lender	Treat as an existing loan and use the transfer message	It implies a loan on the property, which is the transfer trigger.
Plot or land	Ask how it is classified; only agricultural disqualifies	Avoids rejecting residential or commercial plots by assumption.
Qualified lead or loan-transfer case	No live transfer; expert or specialist calls back	The brief says a senior expert will call back shortly and the call then ends.
Market value	Record only	No threshold in the brief; inventing one would be a defect.

6. Tools used and tools deliberately excluded

- **Calendar availability and booking (Cal.com):** when a customer is busy, the agent resolves the requested day and time against the current date and time, offers only slots the calendar returns, confirms the day, date and time aloud, and books a callback. Callbacks must be in the future; times already passed or too soon are refused politely.
- **Transfer Call:** configured only as a safety exit for a customer who insists on speaking to a human immediately, after the agent has explained once that a senior expert will call back. It is never used for qualified, disqualified, busy or loan-transfer cases, so it does not contradict the brief.
- **Web Search (excluded):** the brief forbids invented product facts such as rates. A search tool could return unverified figures, so product answers are restricted to the RAG context.

7. Testing method and results

Each scenario in the assignment brief and reference notes was run as a live call on Bolna, speaking like a real customer (full sentences, fillers, interruptions) in English and Hindi. After each call the transcript was checked against the rule table and the per-turn procedure. **All scenarios below passed.**

#	Scenario	Expected behaviour	Result
1	Fully eligible customer	All 7 items captured, then senior-expert handoff	**PASS**
2	Agricultural property	Immediate polite disqualification, call ends	**PASS**
3	Cash income (self-employed and salaried)	Immediate disqualification	**PASS**
4	Only photocopies of documents	Immediate disqualification	**PASS**
5	Tenure under 3 years	Disqualification	**PASS**
6	Tenure over 15 years	Disqualification	**PASS**
7	Boundary values (75 lakh, 3 years, 15 years)	Continue and qualify	**PASS**

#	Scenario	Expected behaviour	Result
8	Loan above 75 lakh, customer accepts cap	Limit explained; proceeds with 75 lakh	**PASS**
9	Loan above 75 lakh, customer declines	Polite end	**PASS**
10	Joint ownership	Continues; not disqualified	**PASS**
11	Existing loan on the property	Loan-transfer specialist message, call ends, no checklist	**PASS**
12	Request to reduce EMI	Same transfer message, call ends	**PASS**
13	Busy customer	Asks callback time, books a future slot, ends	**PASS**
14	Out-of-order, multi-fact answers	Captured once, never re-asked, next missing item only	**PASS**
15	Self-correction ("10 years, actually 12")	Latest value used	**PASS**
16	Contradiction (residential, then agricultural)	Latest answer wins; disqualify	**PASS**
17	Interest-rate question mid-flow	No invented rate; expert will share; returns to checklist	**PASS**
18	Branch / unrelated question, empty RAG	No fabrication; returns to checklist	**PASS**
19	Interruption mid-question	Brief answer, then resumes	**PASS**
20	Ambiguous answer ("papers are somewhere")	One clarification; no assumption	**PASS**
21	Item skipped after a diversion	Agent returns and asks it before handoff	**PASS**
22	"Who is calling?" at verification	Identifies as AI from the company; re-confirms identity	**PASS**
23	Natural speech with fillers and full sentences	Interpreted correctly	**PASS**
24	Hindi and English runs, agent gender	Configured language and gender held throughout	**PASS**

8. Iteration approach

Prompt changes were made one at a time: run a scenario, note the exact line where behaviour deviated, change a single rule, then rerun that scenario plus the happy path to catch regressions. Rules that matter most (disqualification, transfer, loan cap, handoff gate) are stated as unconditional "never" or "only when" conditions, and the "do not disqualify" cases (joint ownership, loan above cap, depth answers) are stated just as explicitly.

9. Conclusion and next steps

The agent behaves as a rule-accurate, state-aware qualifier: it remembers what the customer has said, applies every eligibility rule consistently, never invents product facts, and does not hand off until all seven points are covered. With more time I would add automated transcript evaluation (an LLM-as-judge against the rule table), more Hindi and Hinglish test variations, and structured outcome logging into a CRM.